November 2, 2023

ASPEN AEROGELS

Q3 2023 FINANCIAL RESULTS CALL

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These risks and uncertainties include, but are not limited to, the following: inability to execute the growth plan through 2025, inability to manage supply chain disruptions to avoid undue delay or impact on operations, inability to create new product, partnership and market opportunities; any sustained downturn in the industry or energy prices; any sustained downturn in the energy, energy infrastructure, chemical and refinery, LNG, sustainable building materials, EV thermal barrier, EV battery materials or other markets; any failure to sustain project-based demand in the subsea, LNG, on-shore or other markets; the right of EV thermal barrier customers to cancel contracts with Aspen at any time and without penalty; any costs, expenses, or investments incurred by Aspen in excess of projections used to develop pricing under the contracts with EV thermal barrier customers; any failure of Aspen or PyroThin to meet contractual specifications and requirements under contracts with EV thermal barrier customers; Aspen's inability to create customer or market opportunities for, including PyroThin; any other battery performance and safety products, battery materials or for other new products developed from Aspen's aerogel technology; any disruption or inability to achieve expected capacity levels in any of our three existing production lines in East Providence, RI or the Mexico assembly facility; any inability to establish or timely establish thermal barrier assembly operations in Mexico or any other location; the failure to receive all regulatory or other approvals required to operate, maintain or expand any of Aspen's facilities; any failure to achieve demand for Aspen's products; any failure to achieve expected price increases or average selling prices for Aspen's products; any significant increase in the cost of raw materials, utilities or any other manufacturing consumable; shortages of raw materials, utilities or any other manufacturing consumable; the failure to generate sufficient operating cash flow or to obtain significant additional capital to pursue Aspen's strategy; the failure of Aspen's products to become widely adopted; the competition Aspen faces in its business; any failure to enforce any of Aspen's patents; any failure to protect or expand Aspen's Aerogel Technology Platform™; any future finding of invalidity of any of Aspen's patents in any jurisdiction; any failure to generate sufficient operating cash flow or to obtain sufficient additional capital to continue to pursue Aspen's new business, technology, patent enforcement, or patent defense strategy; any failure of Aspen's products to meet applicable specifications and other performance, safety, technical and delivery requirements; the general economic conditions and cyclical demands in the markets that Aspen serves; the economic, operational and political risks associated with sales and expansion of operations in foreign countries, including Mexico; the loss of any direct customer, including distributors, contractors and OEMs; compliance with health and safety laws and regulations; the maintenance and development of distribution channels; and the other risk factors discussed under the heading "Risk Factors" in our Annual Report on Form 10-K for the year ended December 31, 2022 and filed with the Securities and Exchange Commission ("SEC") on March 16, 2023, as well as any updates to those risk factors filed from time to time in our subsequent periodic and current reports filed with the SEC. 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Q3 2023 Developments

Record company revenues and continued gross margin expansion driven by accelerating PyroThin® demand

Q3 2023 Highlights

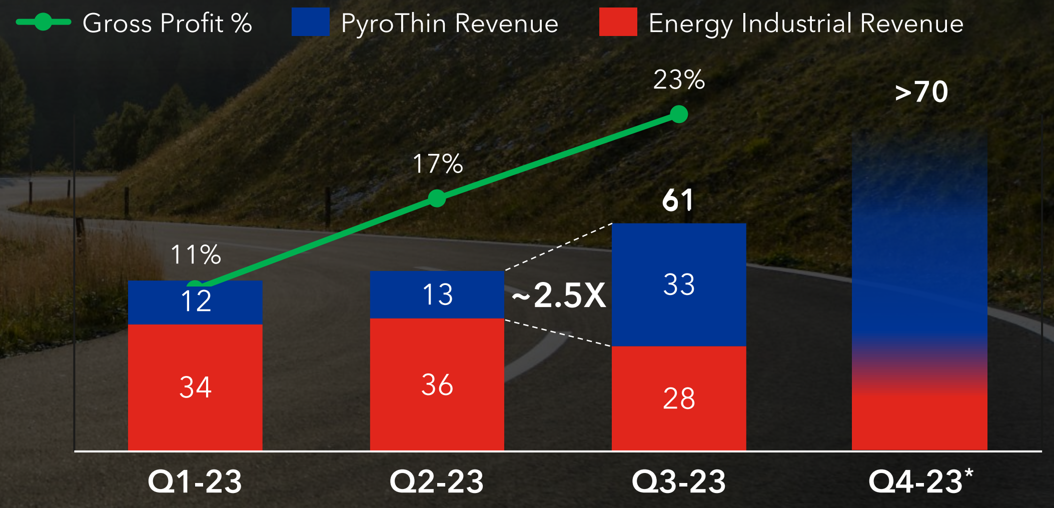
- Record quarterly company revenue of $60.8M
- Record quarterly PyroThin® revenues of $32.8M
- Delivered 23% quarterly gross profit margin
- $118M in unfilled orders for capacity constrained Energy Industrial business

Updated EV OEM Awards

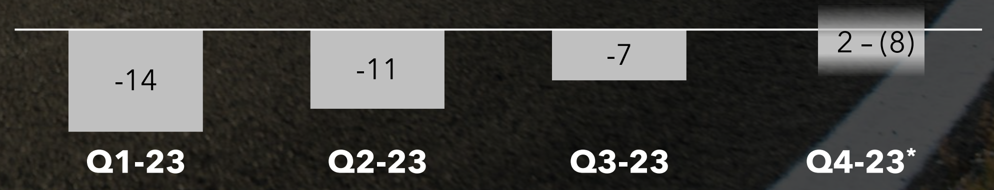
OEM	*Award Year*	*SOP Year*
general motors	*2021*	*2021*
TOYOTA	*2021*	*2021*
SCANIA	*2023*	*2024*
Audi	*2023*	*2025*
TBA	*2023*	
TBA	*2023*	

Financial Performance Progression

2023 Revenue & Gross Margin Trajectory ($ in millions)[1]



2023 Adjusted EBITDA Trajectory ($ in millions)[1,2]



1- Unaudited financials for 2023
2- See slide 12 herein for a reconciliation of net loss, the most directly comparable GAAP measure, to Adjusted EBITDA
* Projected based on current outlook

Q3 and YTD 2023 Financial Highlights

Material costs and fixed cost absorption from higher revenue run rates driving operating loss reductions

$ in millions except per share figures	Q3 2022	Q3 2023[1]	Q3 YTD 2022	Q3 YTD 2023[1]
REVENUES YoY Growth	$36.7 21%	$60.8 66%	$120.8 34%	$154.5 28%
Net (Loss)	$(29.6)	$(13.1)	$(73.1)	$(45.3)
Adj. EBITDA[2]	$(23.2)	$(7.3)	$(56.2)	$(32.1)
EPS	$(0.75)	$(0.19)	$(2.03)	$(0.65)
Cap Ex	$67.0	$32.3	$119.3	$147.7

Commentary

YTD EV Thermal Barrier revenues of $57M, or $33M in Q3, begin to reflect the ramp of GM's EV Thermal Barrier part demand.
YTD Energy Industrial revenues of $98M, or $28M in Q3, reflect supply constrained volume, favorable pricing and a more profitable product mix.

Operating efficiencies from prior investments at our aerogel plant and leverage from moving assembly operations to Mexico drove positive gross margin, while cost discipline in OPEX will contribute to near-term profitability.

Currently optimizing production plans for profitability by manufacturing products that deliver the highest revenue per standard hour of fixed manufacturing costs.

Continued investments to deliver on increasing customer demand.

$10.3M spent towards Plant II in Q3; remainder primarily invested in EV Thermal Barrier assembly tooling in Mexico, along with Rhode Island Aerogel plant improvements, and our labs in the Boston area.

1- Unaudited financials for 2023
2- See slide 12 herein for a reconciliation of net loss, the most directly comparable GAAP measure, to Adjusted EBITDA for the presented period

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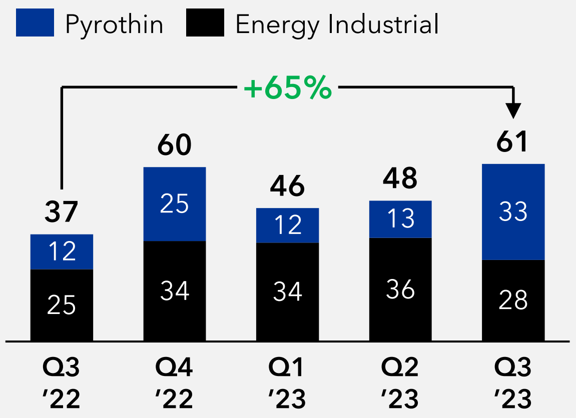
Historical Trends and 2023 Outlook

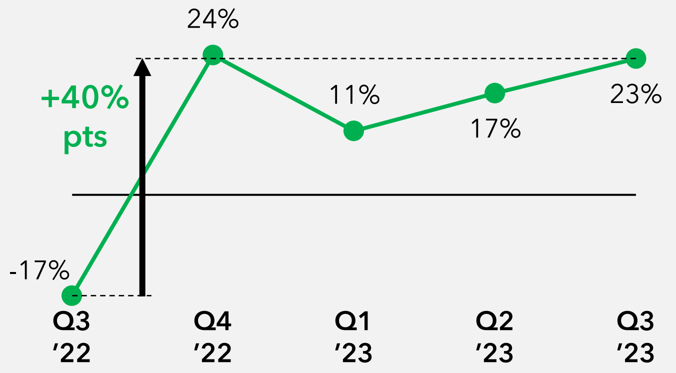
Improved profitability guidance, Q4 performance will be driven by continued increase in EV Thermal Barrier demand

15 Month Quarterly Trajectory[1]

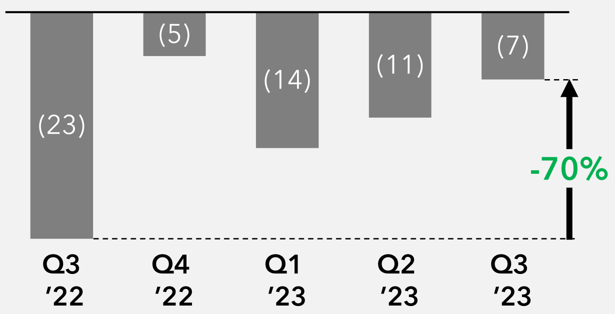
Revenues - $M



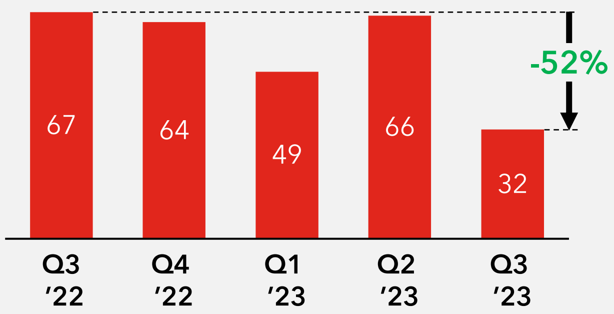
Gross Profit Margin %



Adj. EBITDA[2] - $M



Cap Ex ($M)



2023 Outlook[1]

$ in millions except per share figures

REVENUES YoY Growth *Prior Range*	**>$225** >25% *$200 - $250*
Net (Loss) *Prior Range*	**$(62) - $(52)** *$(85) - $(75)*
Adj. EBITDA[2] *Prior Range*	**$(40) - $(30)** *$(55) - $(45)*
EPS *Prior Range*	**$(0.90) - $(0.76)** *$(1.21) - $(1.07)*
Cap Ex *Prior Range*	**<$175M** *<150M*

1- Unaudited financials for 2023
2- See slide 12 herein for a reconciliation of net loss, the most directly comparable GAAP measure, to Adjusted EBITDA for the presented outlook
Note: Figures may not total due to rounding

Operating Plan Gearing and Recent Results

Results on path towards building a $550M+ revenue business, targeting 25% EBITDA margins with our current asset base

*$ in millions, expressed as **Annual Run Rate***		Q1-23	Q2-23	Q3-23	Q4-23*	LTM Q3-22	LTM Q3-23	LTM % Change
EV Thermal Barrier Revenue	$400M *Capacity*	47	51	131	*Outlook not provided*	36	82	130% revenue growth
Energy Industrial Revenue	*Capacity* $150M	136	142	112		116	132	13% revenue growth
Total Annual Revenue	$550M	182	193	244	>280	152	214	41% revenue growth
Material Costs[1]	40-45%	41%	36%	36%	*Outlook not provided*	55%	38%	17% of sales improvement
Mfg. Costs[1]	20-25%	48%	46%	41%		52%	42%	10% of sales improvement
Potential Gross Profit[1]	35% 200	11%	17%	23%		(7%)	19%	26% of sales improvement
OPEX[1]	<20% Up to 110	53%	53%	47%		49%	48%	1% of sales improvement
Potential EBITDA[1]	25% 140	-56	-43	-29	8 - (32)	-68	-37	47% loss reduction

1. *Numbers are a percentage of total revenue*
* Projected based on current outlook
Note: Figures may not total due to rounding

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Summary

Earning our right to win by leveraging the Aerogel Technology Platform®, executing, and investing to grow profitably

ACCELERATING NEAR TERM PROFITABILITY

Leveraging existing assets and agreements to deliver $550M annual revenue capacity, $200M gross profit, and $140M EBITDA

INCREASING FLEXIBILITY TO MEET DEMAND

Executing supplemental supply agreement to fulfill increasing demand for capacity constrained Energy Industrial business

ADVANCED PYROTHIN® COMMERCIAL PROGRESS

Continued traction towards diversified customer base with 4 current PyroThin® awards; on track for 6 total awards by year-end

DEMONSTRATED OPERATING EXECUTION

Q3 saw record level PyroThin revenues of $32.8M, while company gross profit of 23% highlights improving margin progression

The Road to Creating Near Term Value

- Start Energy Industrial Supplemental Supply
- Secure Additional OEM Thermal Barrier Awards
- Deliver EV Thermal Barrier Demand Ramp
- Optimize OPEX and Hold it Flat
- Reduce Fixed Conversion Costs
- Minimize and Right-time Near Term CAPEX

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Appendix

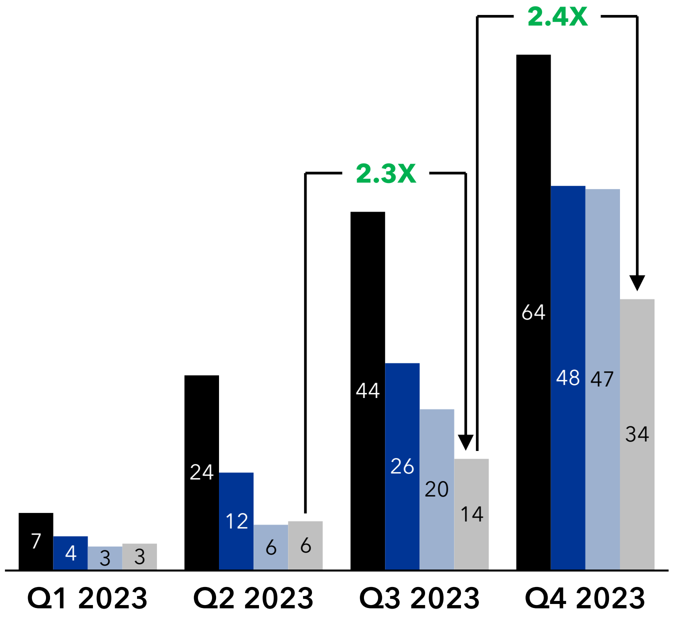
Demand Overview - GM Ultium IHS Forecast

GM Ultium production doubled from Q2 to Q3; ramp projected to continue into Q4 and 2024

Ultium platform volumes are accelerating heading into Q4

2023 GM Ultium EV Production Forecast[1]
(Vehicles in thousands)

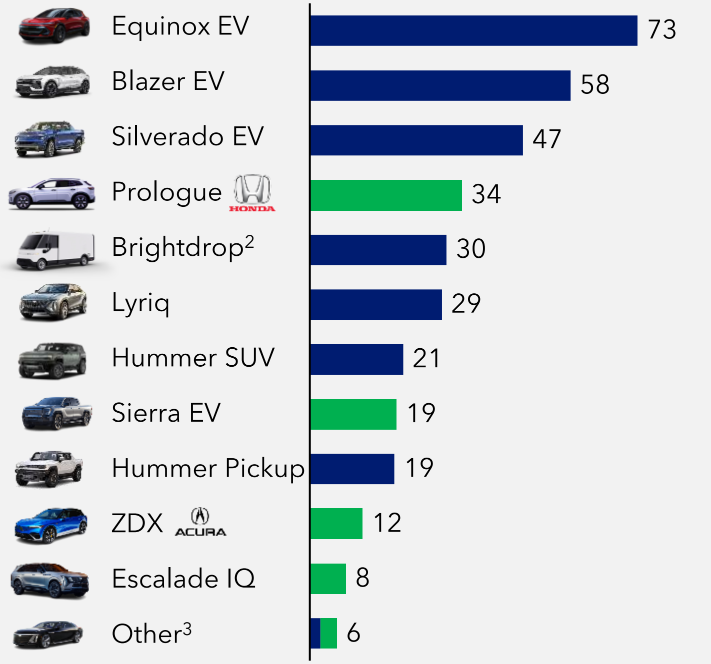
- Q1 '23 Forecast (Total – 140k)
- Q2 '23 Forecast (Total – 90k)
- Q3 '23 Forecast (Total – 76k)
- Q4 '23 Forecast (Total – 57k)



2024: major GM nameplates to scale; new Honda launches

2024 GM Ultium EV Production Forecast[1]
(Vehicles in thousands from Q4 '23 forecast)

- In-Production (2023)
- 2024 Launch

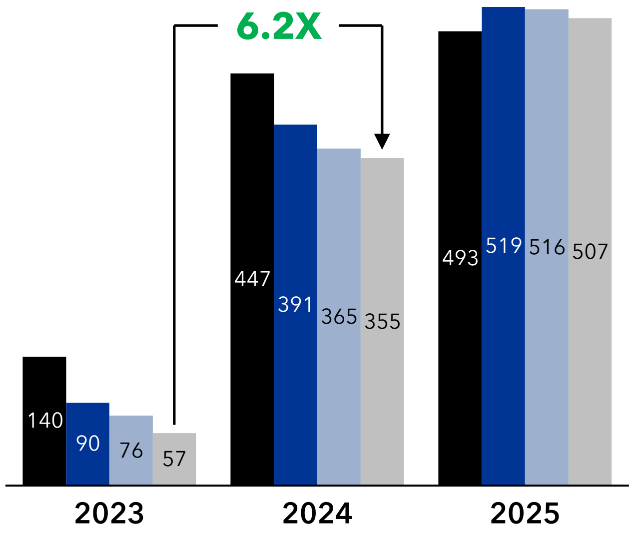


The long-term outlook remains strong

GM Ultium EV Production Forecast[1]
(Vehicles in thousands)

- Q1 '23 Forecast
- Q2 '23 Forecast
- Q3 '23 Forecast
- Q4 '23 Forecast

Note: Forecast does not include impacts of recently announced Orion assembly delay (IHS with ~66k units in 2025)



1- IHS January 2023, April 2023, July 2023, October 2023 vehicle production forecasts
2. Brightdrop includes Zevo 400 and Zevo 600
3. Other includes Optiq, Origin, and Celestiq

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GM North American Ultium Battery Manufacturing

Plant Location	Battery Partner	Estimated Cost	Status	Expected SOP	Capacity	Notes
Lordstown, OH	LG Chem	$2.3B	Completed	In-production (2022)	30 GWh	Testing high-capacity module assembly lines; Expected to reach full capacity in November (targeting 36M cells in 2023 and 100M cells in 2024)
Spring Hill, TN	LG Chem	$2.3B	Under Construction	Early 2024	50 GWh	Expected to create 1,300 jobs and supply GM's Spring Hill assembly plant
Lansing, MI	LG Chem	$2.6B	Under Construction	Late 2024	50 GWh	Will be 2.8M sq ft and employ ~1,700 workers
New Carlisle, IN	Samsung	$3B	Announced	2026	30 GWh	Expected to create ~1,700 jobs; construction yet to begin

Source: Public information (i.e., company press releases, announcements, and communications)

GM North American Ultium EV Assembly Plants

Assembly	Union Status	Est 2024 Production[1]	Est 2025 Production[1]	SOP[1]	Nameplates	Est 2024 Production[1]	Est 2025 Production[1]	Notes
Ramos, Mexico	CTM	168k	191k	2023	Equinox EV	73k	83k	Also builds ICE variants of the Blazer and Equinox
				2023	Blazer EV	58k	54k	
				2024	Prologue	34k	39k	
				2024	Optiq	3k	15k	
Factory Zero, Michigan	UAW	114k	164k	2023	Silverado EV	45k	80k	Fully dedicated to EV assembly
				2023	Hummer EV	21k	19k	
				2024	Sierra EV	19k	30k	
				2021	Hummer EV Pickup	19k	17k	
				2024	Escalade IQ	8k	11k	
				2023	Origin	2k	3k	
				2025	Escalade IQL	-	4k	
Spring Hill, Tennessee	UAW	41k	49k	2022	Lyriq	29k	27k	Also builds ICE vehicles
				2024	ZDX	12k	17k	
				2025	Symboliq	-	5k	
CAMI II, Canada	Unifor	30k	36k	2023	ZEV 400	17k	22k	In October announced production pause through Spring 2024 due to delays in battery-module supply
				2023	ZEV 600	13k	15k	
Orion, Michigan	UAW	2k	66k	2024	Silverado	2k	53k	Assembly SOP moved to late 2025, however both nameplates are also produced at Factory Zero
				2024	Sierra EV	<1k	13k	

1- IHS October 2023 vehicle production forecast, figure does not include Celestiq nameplate with <1k units assembled in Warren Technical Center
Note: Figures may not total due to rounding
Sources: IHS, Public information (i.e., company press releases, announcements, and communications)

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GAAP to Non-GAAP Reconciliation

($ in thousands)	Q3 2023	Q3 2022	Q3 YTD 2023	Q3 YTD 2022	FY 2023 Outlook Low	FY 2023 Outlook High
Net Loss	(13,073)	(29,595)	(45,292)	(73,129)	(62,400)	(52,400)
Depreciation and Amortization	4,550	2,531	10,757	6,692	15,300	15,300
Stock-based Compensation	2,789	2,590	7,766	6,713	10,400	10,400
Other (Income) Expense	(1,561)	1,286	(5,294)	3,550	(3,300)	(3,300)
Adjusted EBITDA	**(7,295)**	**(23,188)**	**(32,063)**	**(56,174)**	**(40,000)**	**(30,000)**

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